

[2]



$$(1) \quad m \cdot F = k(S_2 - S_1) - kS_1 = kS_2 - 2kS_1 \quad (5)$$

$$3m \cdot F = -kS_2 - k(S_2 - S_1) = -2kS_2 + kS_1 \quad (5)$$

$$\begin{cases} m \cdot \frac{d^2 S_1}{dt^2} = -2kS_1 + kS_2 & - (1) \quad \text{10} \quad (5) \\ 3m \cdot \frac{d^2 S_2}{dt^2} = kS_1 - 2kS_2 & - (2) \quad \text{10} \quad (5) \end{cases}$$

(2) 가장 간단한 수학적 조합

$$\rightarrow S_1 = A \cos(\omega t), \quad S_2 = B \cos(\omega t) \quad (5)$$

$$(1); \quad m \cdot (-\omega^2 A \cos(\omega t)) = -2kA \cos(\omega t) + kB \cos(\omega t)$$

$$(2k - m\omega^2)A = kB \quad - (3)$$

$$(2); \quad 3m \cdot (-\omega^2 B \cos(\omega t)) = kA \cos(\omega t) - 2kB \cos(\omega t)$$

$$-kA = (3m\omega^2 - 2k)B \quad - (4)$$

$$(3) \div (4)$$

$$\frac{2k - m\omega^2}{-k} = \frac{k}{3m\omega^2 - 2k}$$

$$-3m^2\omega^4 + 6mk\omega^2 - 4k^2 = -k^2$$

$$3m^2(\omega^2)^2 - 6mk\omega^2 + 3k^2 = 0$$

$$\omega^2 = \frac{6mk \pm \sqrt{64m^2k^2 - 36m^2k^2}}{6m^2}$$

$$= \frac{6mk \pm 2mk\sqrt{7}}{6m^2} = \frac{k}{m} \cdot \frac{4 \pm \sqrt{7}}{3} \quad (5)$$